

Question Number	Answer	Additional guidance	Mark
7(a)	An explanation that makes reference to three of the following points: • all introns {cut out / removed} (1) • some exons {cut out / removed} (1) • by {spliceosomes / enzymes} (1) • exons are joined together to form active mRNA (1)	REJECT some introns removed IGNORE exons removed unqualified ACCEPT splicing / pre-mRNA spliced ACCEPT exons can be {rearranged / joined together in different order} IGNORE coding regions	(3)

Question Number	Answer	Mark
7(b)(i)	The only correct answer is C rough endoplasmic reticulum <i>A is not correct because the Golgi apparatus does not have ribosomes on its surface</i> <i>B is not correct because mitochondria do not have ribosomes on its surface</i> <i>D is not correct because smooth endoplasmic reticulum does not have ribosomes on its surface</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	An answer that makes reference to the following point: (site of) {translation / polypeptide synthesis} (1)	ACCEPT {produce / make} {polypeptides / 1° structure} ACCEPT form {chains / sequence} of amino acids IGNORE proteins	(1)

Question Number	Answer	Mark
7(b)(iii)	The only correct answer is B polypeptide and RNA strand <i>A is not correct because Z is RNA</i> <i>C is not correct because X is a polypeptide</i> <i>D is not correct because X is a polypeptide</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	An answer that makes reference to the following points: - binds {methyl group / CH₃ molecule} to DNA (1) - to {cytosine / CpG} (in promoter region) (1)	REJECT (occurring on) histones / amino acids IGNORE {removing / bringing} methyl groups REJECT cysteine	(2)

Question Number	Answer
*7(c)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • over the 21 days the tumour volume increased in all four groups • there was the greatest increase in tumour volume in group A / lowest increase in group D • group D increased in a similar way to group B and C until day 13 / group D decreases in volume after day 15 • higher concentration of peperomin E extract resulted in a smaller tumour mass • after 21 days there was the {greatest mean tumour mass in group A / untreated group} / after 21 days there was the {lowest mean tumour mass in group D / current treatment} • {current drug / D} most effective / converse • higher concentration of peperomin E extract is more effective than lower / converse • tumour {increases / formation} due to {mitosis / uncontrolled cell division} • higher mitotic index in group A / converse • {peperomin E extract / current drug} reduces mitotic rate • {peperomin E extract / drug(s)} {removes methyl groups on gene / is involved in demethylation / inhibits DNMT} • suggestions of method of action for current drug • demethylation allows {transcription factors / RNA polymerase} to bind / DNA uncoils / euchromatin / no supercoiling eq • {RASSF1 / tumour suppressor} gene is {activated / switched on} • {RASSF1 / tumour suppressor} gene is transcribed / translation occurs / {RASSF1 / tumour suppressor} protein synthesised • {RASSF1 / tumour suppressor gene} protein slows down {cell division / growth of tumours} / suitable suggestions of function of tumour suppressor protein e.g. disruption of spindle fibres (ignore gene slows cell division etc) • no information on starting mass of tumour / do not know if the mass of tumour increased with current drug • suitable comments on methodology of investigation e.g. only done on lung cancer tumours / short time scale / don't know long term effect / no comparison of side effects from different drugs / no SD bars so unsure if significant difference between groups / small numbers of mice / no information on controlled variables / overlapping of volume lines for the 2 peperomin concentrations / no current drug concentration information (6)

		Additional guidance
Level 0	0	No awardable content
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and / or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and / or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.